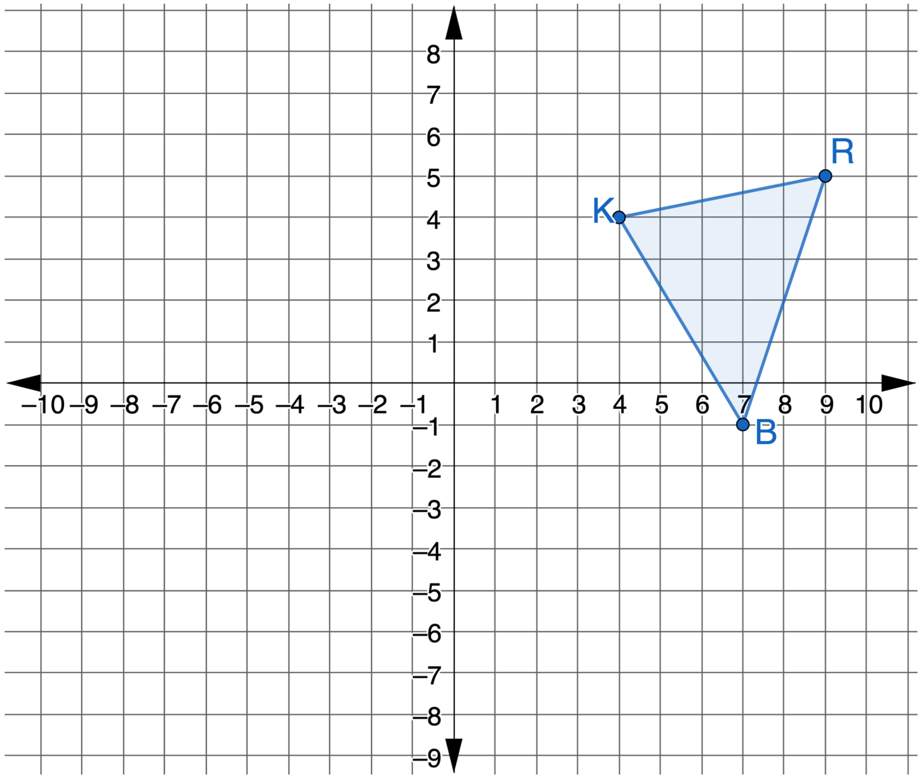
4.5.5 Wrap-Up: Ticket Out the Door

1. Triangle KRB is shown on the coordinate grid.



Perform the sequence of transformations on ΔKRB. Assume all rotations are centered at the origin. Label the result of the transformations as ΔTLS.

- $(x, y) \to (x - 12, y + 3)$
- $(x, y) \to (x, -y)$

Unit 4, Lesson 6: Isometry

Benchmarks

MA.912.GR.2.3 Specify a sequence of transformations that will map a given figure onto itself, or onto another congruent or similar figure.

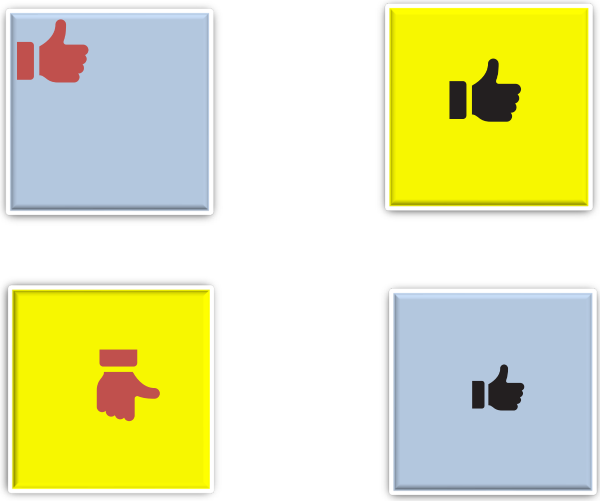
MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.

Learning Targets

- ☐ I can describe a sequence of transformations that map one figure onto another congruent figure.
- ☐ I can draw an image given a preimage and a sequence of transformations on a coordinate plane.

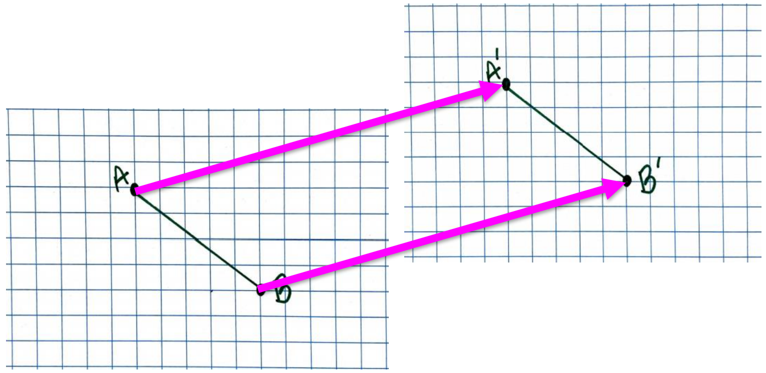
4.6.1 Warm-Up: Which One Doesn't Belong?

1. Which one doesn't belong? Explain your choice.



4.6.2 Exploration: Transforming Parts

1. Traci drew $\overline{AB}$ on a piece of graph paper. She then moved the graph paper up and over to the right, as shown by the arrows. The movement created $\overline{A'B'}$.



The graph paper is a real-life example of a plane. When an object is moved, flipped, or rotated, it is the plane that the object is on that moved. This means that every point on the plane is moved, flipped, or rotated.

a. Draw $\overline{YZ}$ on the same plane as $\overline{AB}$. Then, draw $\overline{Y'Z'}$ on the second plane to show how $\overline{YZ}$ would move in the same way as $\overline{AB}$ when the plane is moved.

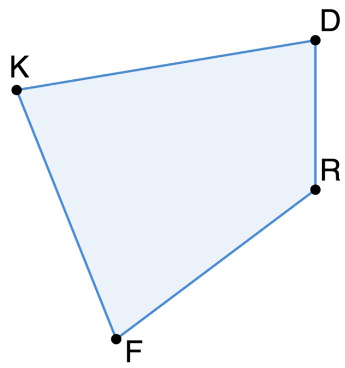
b. Draw the arrows from the endpoints of $\overline{YZ}$ to the endpoints of $\overline{Y'Z'}$. Compare the arrows from $\overline{YZ}$ to $\overline{Y'Z'}$ to the arrows from $\overline{AB}$ to $\overline{A'B'}$. What do you notice?

Isometry is a transformation that preserves distance and angle measure.

c. Use the definition of isometry to justify that Traci's example is an isometry.

4.6.3 Exploration: Transformations and Isometry

1. Quadrilateral $KDRF$ is shown.



a. Use patty paper to copy the length of $\overline{RF}$. Label the endpoints on the patty paper. Flip the patty paper and place it in the space to the left of quadrilateral $KDRF$. Use the patty paper and any other tools to draw the copy of $\overline{RF}$. Name it $\overline{R'F'}$.

A ***transformation*** is a one-to-one correspondence between two sets of points.

b. Work with your partner to complete the statement:

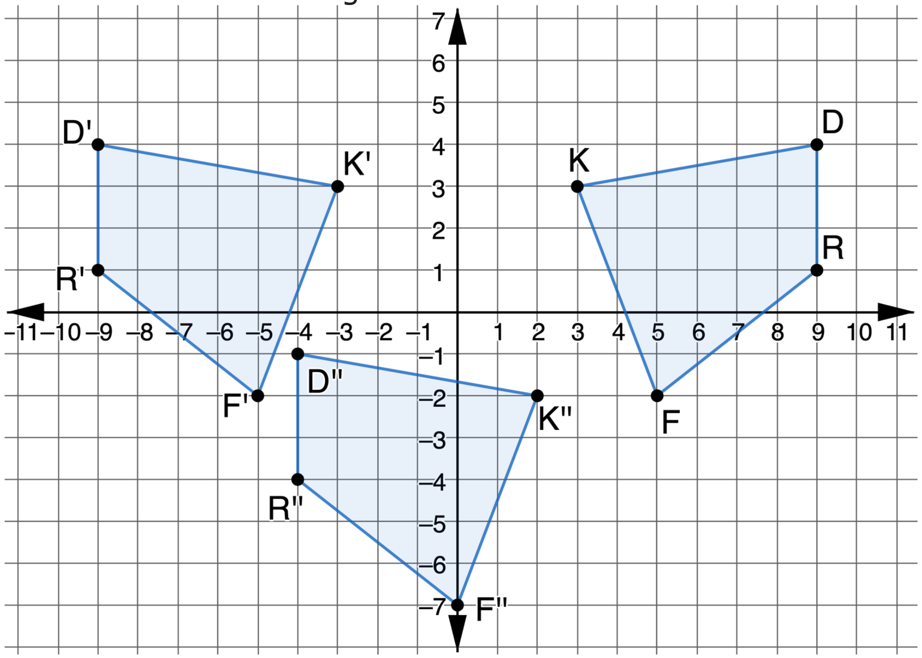
Since there are a set of points on $\overline{RF}$ and a set of points on $\overline{R'F'}$, it is not enough to just state that the endpoints R and F correspond with endpoints R' and F' because a transformation is a ________________ ________________________ between two sets of points. Therefore, to show that $\overline{RF} \cong$ ______, all of the points on $\overline{RF}$ have to have a matching point on $\overline{R'F'}$.

c. Use the steps to apply the same transformations to each part of quadrilateral $KDRF$.

- Use patty paper to copy $\angle RFK$. Use the copy of $\angle RFK$ to create $\angle R'F'K'$.
- Use the patty paper and any other tools to draw the copy of $\overline{KF}$. Name it $\overline{K'F'}$.
- Use patty paper to copy $\angle FRD$. Use the copy of $\angle FRD$ to create $\angle F'R'D'$.
- Use the patty paper and any other tools to draw the copy of $\overline{RD}$. Name it $\overline{R'D'}$.
- Use patty paper to copy $\angle FKD$. Use the copy of $\angle FKD$ to create $\angle F'K'D'$.
- Use the patty paper and any other tools to draw the copy of $\overline{KD}$. Name it $\overline{K'D'}$.

d. Explain why when drawing all of the parts of the quadrilateral $KDRF$ to create quadrilateral $K'D'R'F'$, the movements of the patty paper were the same as when drawing $\overline{R'F'}$.

2. Quadrilaterals $KDRF$, $K'D'R'F'$, and $K''D''R''F''$ are shown on the coordinate grid.



a. Use patty paper to copy quadrilateral $KDRF$.

b. Lay the copy of quadrilateral $KDRF$ on top of quadrilateral $KDRF$.

c. Describe what should be done so that the copy of quadrilateral $KDRF$ lays on top of quadrilateral $K'D'R'F'$.

d. Was the movement of the patty paper an example of isometry?

e. Lay the copy of quadrilateral $KDRF$ on top of quadrilateral $K'D'R'F'$.

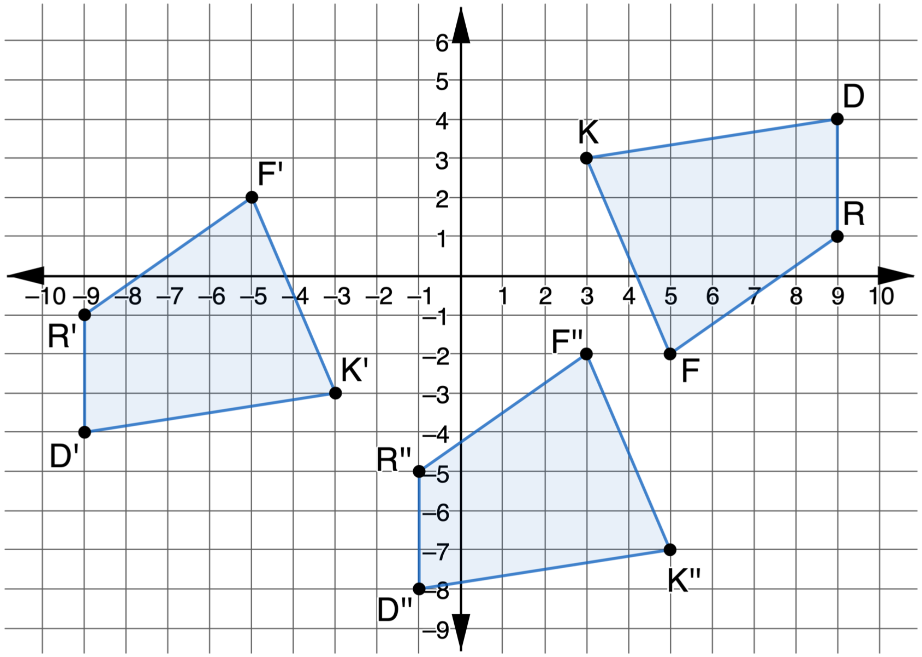
f. Describe what should be done next so that the copy of quadrilateral $KDRF$ lays on top of quadrilateral $K''D''R''F''$.

g. Was the movement of the patty paper an example of isometry?

h. Determine if the copy of quadrilateral $KDRF$ could be moved with just one transformation so that it lays on top of quadrilateral $K''D''R''F''$.

3. There is at least one other sequence of transformations that can be used to move the copy of quadrilateral $KDRF$ so that it lays on top of quadrilateral $K''D''R''F''$. Work with your partner to describe one of these sequences of transformations.

4. The coordinate grid shows a different sequence of transformations of quadrilateral $KDRF$.



Describe what movements should be done with your patty paper copy of quadrilateral $KDRF$ so that the copy of quadrilateral $KDRF$ lays on top of quadrilateral $K''D''R''F''$, by first laying on top of quadrilateral $K'D'R'F'$.

4.6.4 Bring It Together: Definition of Congruence in Terms of Rigid Motions

1. For both sequences of transformations of quadrilateral

$KDRF$, the paper copy of quadrilateral $KDRF$ was

- ○ congruent
- ○ not congruent

to

- ○ $K'D'R'F'$ only.
- ○ $K''D''R''F''$ only.
- ○ both $K'D'R'F'$ and $K''D''R''F''$.

The transformations used in both sequences were

- ○ dilations.
- ○ rigid motions.
- ○ dilations and rigid motions.

All of the points of

- ○ $K'D'R'F'$
- ○ $K''D''R''F''$
- ○ $K'D'R'F'$ and $K''D''R''F''$

can be mapped to all

of the points of $KDRF$.

Definition of Congruence in Terms of Rigid Motions

Two figures are congruent if and only if there exists one or more rigid motions, which will map one figure onto the other.

2. Use the definition of congruence in terms of rigid motions to explain how both sequences of transformations for quadrilateral $KDRF$ resulted in $KDRF \cong K''D''R''F''$.

4.6.5 Wrap-Up: Growth Mindset

The power of "YET."

"The power of believing that you can improve." - Carol Dweck

Select all of the concepts that apply to your understanding right now.

1. I don't understand
   - ○ isometry
   - ○ transformations
   - ○ the definition of congruence in terms of rigid motions

   YET.

2.
   - ○ Isometry
   - ○ Mapping
   - ○ One-to-one correspondence
   - ○ Definition of congruence in terms of rigid motions

   doesn't make sense to me YET.

3. I can explain
   - ○ isometry.
   - ○ mapping.
   - ○ one-to-one correspondence.
   - ○ transformations.
   - ○ the definition of congruence in terms of rigid motions.

Unit 4, Lesson 7: Justifying Congruence

Benchmarks

MA.912.GR.2.3 Specify a sequence of transformations that will map a given figure onto itself, or onto another congruent or similar figure.

MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.

Learning Targets

- ☐ I can describe a sequence of transformations that will map a preimage and an image to show congruence.
- ☐ I can describe a sequence of transformations that map one figure onto another congruent figure.